EE582 Homework 01 Report

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Question 1

A shunt harmonic filter (series RLC) is shown in Fig. 1(a). It is designed for a three-phase 13.8 kV, 60 Hz system to shunt the 7th harmonic current. The circuit parameters are:

- $r_L = 1\ \Omega$
- $L = 19$ mH
- $C = 8.2\ \mu$F

Step 1: Write the KVL equation:

$$V_{\text{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L\frac{di_L}{dt} + r_L i_L + v_C = V_{\text{dc}}$$

With $i_L = C\frac{dv_C}{dt}$ and $v_C = \frac{1}{C}\int i_L dt$.

Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L}\frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19\times10^{-3}\times 8.2\times10^{-6}}} = 2533.5 \text{ rad/s}$$
$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19\times10^{-3}}{8.2\times10^{-6}}}} = 0.0104$$
$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2533.3 \text{ rad/s}$$

Step 5: Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5 \times 0.019}{1} = 48.136$$

Step 6: Eigenvalues:

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Step 7: Reactive power:

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\boxed{\omega_n = 2533.5 \text{ rad/s}}$
- Damping ratio: $\boxed{\zeta = 0.0104}$
- Damped frequency: $\boxed{\omega_d = 2533.3 \text{ rad/s}}$
- Quality factor: $\boxed{Q = 48.136}$
- Eigenvalues: $\boxed{-26.316 \pm 2533.3j}$
- Reactive power: $\boxed{-602.04 \text{ kVAr}}$

Question 2

Consider the series RLC circuit with parameters r_L, L, and C as in Q1. The equations are:

KVL:

$$V_{\text{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0 \tag{1}$$

where i_L is the inductor current and v_C is the capacitor voltage.

Capacitor current:

$$i_L = C\frac{dv_C}{dt} \tag{2}$$

Substitute (2) into (1):

$$V_{\text{dc}} - r_L C\frac{dv_C}{dt} - LC\frac{d^2v_C}{dt^2} - v_C = 0$$

Rearrange to standard form:

$$LC\frac{d^2v_C}{dt^2} + r_L C\frac{dv_C}{dt} + v_C = V_{\text{dc}}$$

Divide both sides by LC:

$$\frac{d^2v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{V_{\text{dc}}}{LC}$$

This is a standard second-order nonhomogeneous ODE. The homogeneous solution is:

$$\frac{d^2v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = 0$$

Eigenvalue computation: The characteristic equation is:

$$s^2 + \frac{r_L}{L}s + \frac{1}{LC} = 0$$

The eigenvalues are:

$$s = \frac{-\frac{r_L}{L} \pm \sqrt{\left(\frac{r_L}{L}\right)^2 - 4\cdot\frac{1}{LC}}}{2}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F):

$$\frac{r_L}{L} = \frac{1}{0.019} = 52.63\ \text{s}^{-1}$$
$$\frac{1}{LC} = \frac{1}{0.019 \times 8.2 \times 10^{-6}} = 6,430,041\ \text{s}^{-2}$$

Plug into the quadratic formula:

$$s = \frac{-52.63 \pm \sqrt{(52.63)^2 - 4 \times 6,430,041}}{2}$$

The discriminant is negative, so the roots are complex conjugates:

$$s = -26.316 \pm j\,2533.3$$

Interpretation: The eigenvalues are complex conjugates with negative real parts, confirming the circuit is underdamped. Based on the very low value of the damping ratio $\zeta = 0.0104$ calculated in Q1, the transient response will exhibit oscillatory behavior with a slow decay.

General solution:

$$v_C(t) = v_{C,\infty} + [v_{C,0} - v_{C,\infty}]\left[1 - \frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_d t + \phi)\right]$$

where $v_{C,\infty}$ is the steady-state value, $v_{C,0}$ is the initial value, and $\omega_d = \omega_n\sqrt{1-\zeta^2}$.

For a step input, $v_{C,\infty} = x_{SS}$, and the solution simplifies to:

$$\boxed{x(t) = x_{SS}\left[1 - \frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_d t + \phi)\right]}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F), the numeric solution is:

$$\boxed{x(t)[kV] = 11.3 * \left[1 - 1.00005\, e^{-26.34\, t}\, \sin(2533.3\, t + 1.5604)\right]}$$

Question 3

Develop a Simulink model of the circuit shown in Fig. 1(a) using conventional Simulink blocks (i.e., summers, gains, integrals, etc.). Compute the transient response (solutions) on time interval $[0, 0.1]$ s using the ode3 solver with 0.1 ms time-step size.

Block Diagram

The following block diagram represents the Simulink implementation of the RLC circuit for part (a):

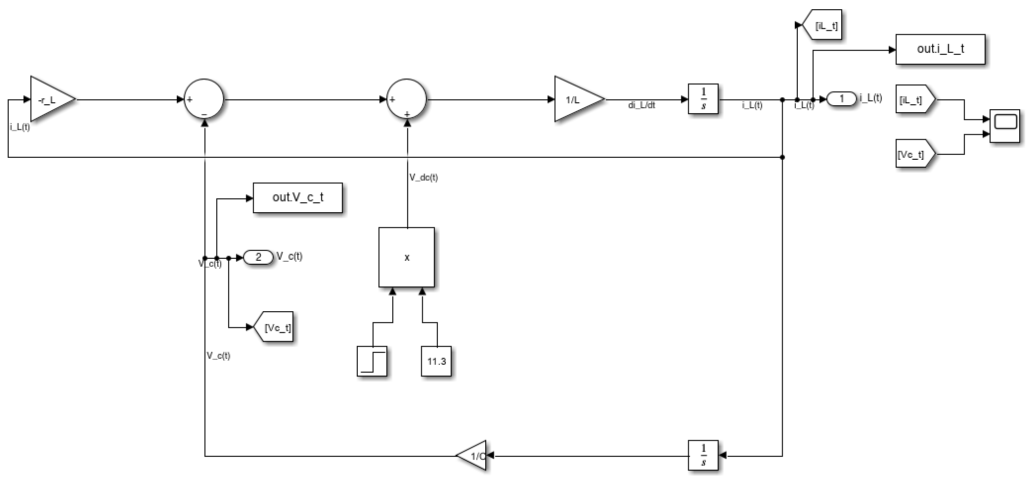

Figure 1: Simulink block diagram for the series RLC circuit (part a).

Simulation Results

The transient response of the circuit was simulated in Simulink using the `ode3` solver with a fixed time step of 0.1 ms over the interval $[0, 0.1]$ s. The following plot shows the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$:

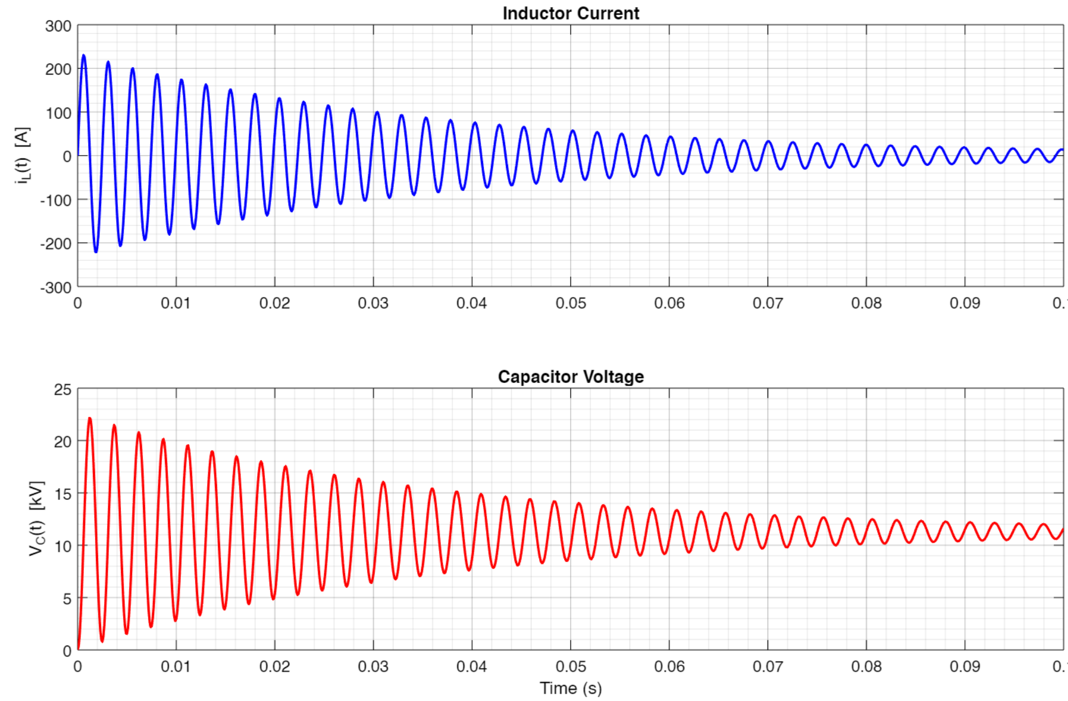

Figure 2: Simulink simulation results: $i_L(t)$ and $v_C(t)$ for the RLC circuit.

Discussion

The simulation results show the expected underdamped oscillatory response, consistent with the analytical solution derived in Q1 and Q2. The current and voltage waveforms exhibit oscillations with a slow decay, as predicted by the low damping ratio ($\zeta = 0.0104$). The Simulink model accurately captures the circuit dynamics using only basic blocks and the specified solver settings.

Question 4

Question:
Find the system eigenvalues using the MATLAB functions `linmod` and `eig` on the command line. Compare the results with part 2.

MATLAB Output

The following figure shows the eigenvalues computed using MATLAB's `linmod` and `eig` functions:

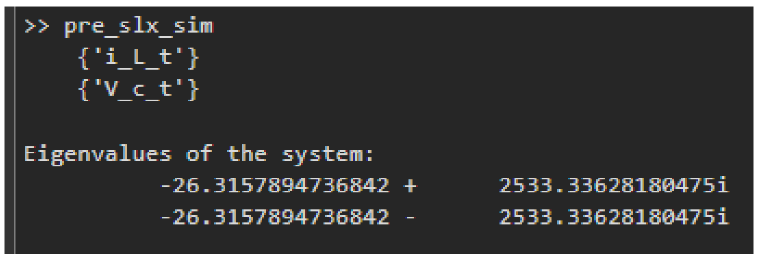

Figure 3: Eigenvalues of the system computed using `linmod` and `eig` in MATLAB.

Comparison with Analytical Results

The eigenvalues obtained from MATLAB are:

$$-26.3158 \pm 2533.34i$$

These match very closely with the hand-computed values in Q2:

$$-26.316 \pm 2533.3i$$

The small numerical difference is due to rounding and floating-point precision in MATLAB. Both methods confirm the system is underdamped with complex conjugate eigenvalues, as expected from the analytical derivation.

Question 5

Analyze the stability of the Forward Euler (ode1) and explicit RK5 (ode5) methods for the RLC circuit. Plot the stability regions and indicate the range of h for which the methods are stable.

Stability Plots

The following figure shows the stability regions for Forward Euler (ode1) and explicit RK5 (ode5) methods, with the scaled eigenvalues $z = h\lambda$ plotted for various step sizes h:

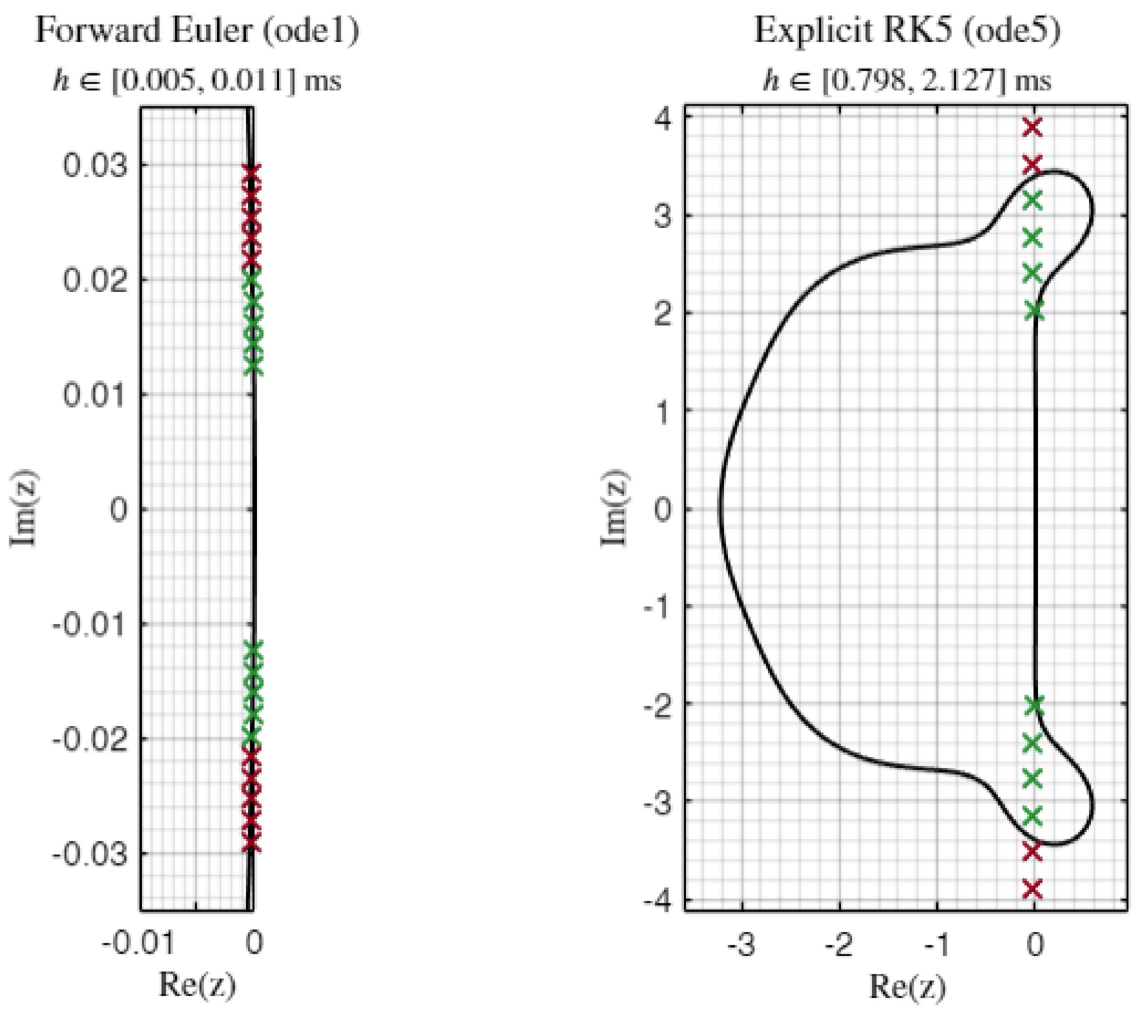

Figure 4: Stability regions for Forward Euler (ode1, left) and explicit RK5 (ode5, right). Each cross represents $z = h\lambda$ for a candidate step size h and the system eigenvalues λ. Green crosses are inside the stability region (stable), red crosses are outside (unstable). Note that the plot for Forward Euler is zoomed in to show the small stability region.

Stability Criteria and z Formulae

For a linear system $\dot{x} = \lambda x$, the stability of a numerical method is determined by the location of $z = h\lambda$ in the method's stability region:

- **Forward Euler (FE):** The stability function is $R(z) = 1 + z$. The method is stable if $|1 + z| < 1$.
- **Explicit RK5 (Taylor-5):** The stability function is $R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!}$. The method is stable if $|R(z)| < 1$. (Note: MATLAB's ode5 is not exactly a Taylor-5 method, but the Taylor-5 region is a good approximation for illustration.)

Interpretation of Crosses

Each cross in the plot corresponds to a value of $z = h\lambda$ for a particular step size h and one of the system's eigenvalues λ.
Green crosses indicate z values inside the stability region (method is stable for that h), while **red crosses** are outside (unstable for that h).

Maximum Stable Step Size

From the plot and MATLAB output:

- **Euler:** $h_{\max} \approx 0.008$ ms
- **Taylor-5:** $h_{\max} \approx 1.329$ ms

These values indicate the largest time step for which each method remains stable for the given system.

Summary

The explicit RK5 (Taylor-5) method allows for a much larger stable time step than Forward Euler, as seen from the much larger stability region. The actual ode5 implementation in MATLAB is not exactly Taylor-5, but the qualitative stability behavior is similar for this analysis.

Question 6

Use any one of the fixed-time-step solvers of your own choice. Plot the solution for i_L and v_C. Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust. Compare your result with the result of using the ode5 at $h = 0.5 \times 10^{-3}$ s and $h = 1.0 \times 10^{-3}$ s.

Simulation Results

The following figures show the results of the fixed-step simulations:

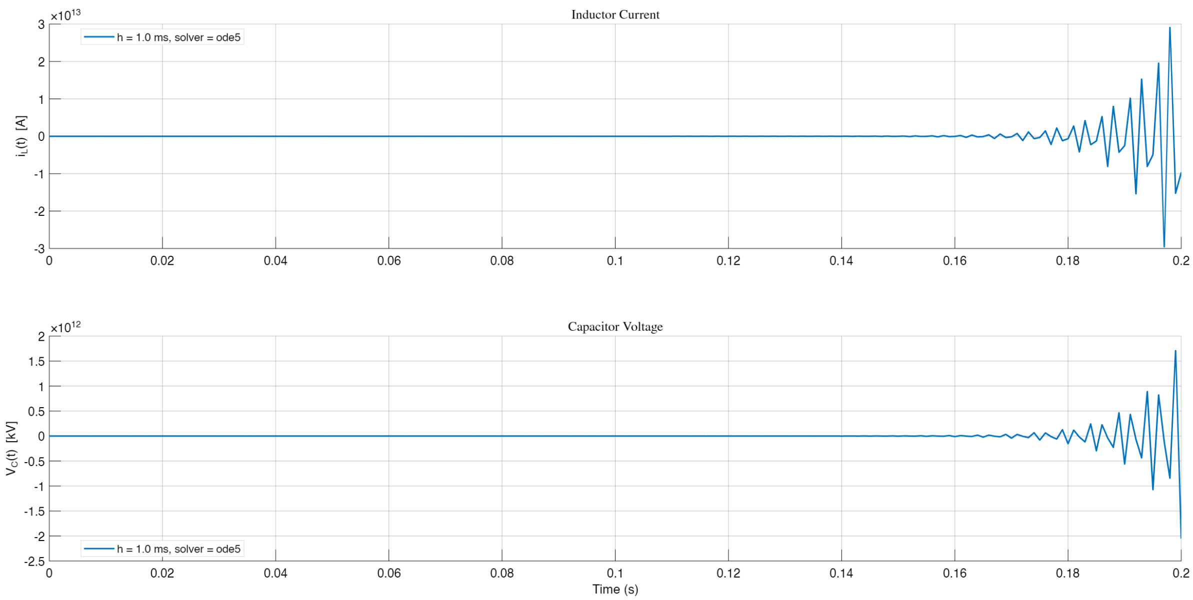

Figure 5: Diverging solution for $h = 1$ ms using ode5 (RK5).

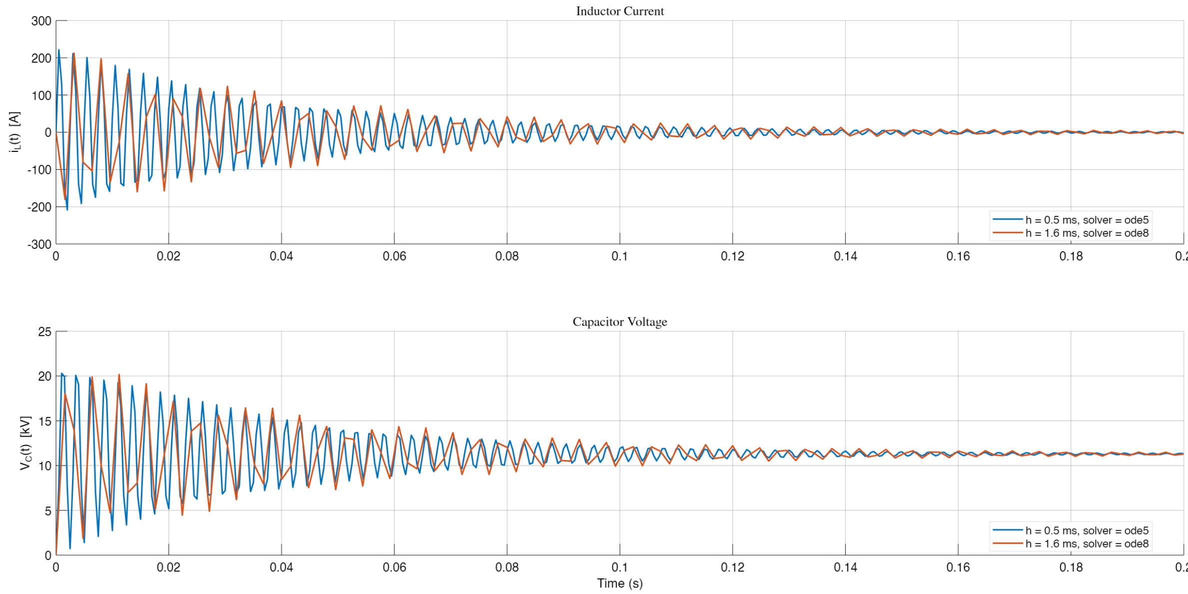

Figure 6: Converging solutions for RK8 with $h = 1.6$ ms and ode5 with $h = 0.5$ ms. The legend indicates the solver and step size.

Discussion

For this part, I used an explicit RK8 method for my own solver. The maximum time step $h_{\max}$ for which the RK8 method produced a stable and accurate solution was determined experimentally to be 1.6 ms. For ode5 (RK5), the $h = 1$ ms case is shown separately above, as it diverges and would spoil the scale of the converging plot.

The experimentally obtained $h_{\max}$ for RK8 was found to be less than the theoretical value for Taylor-5 (≈ 1.3 ms). For ode5 (RK5), the maximum stable h was found to be less than 0.7 ms (figure not shown here), which is also less than the theoretical prediction. The theoretical values are based on the Taylor-5 stability region, but ode5 is not exactly Taylor-5, and RK8 has a different (larger) stability region.

Remark

The fact that the experimentally determined $h_{\max}$ is smaller than the theoretical value is mainly due to (i) the difference between the actual Dormand–Prince implementation of `ode5` and the idealized Taylor-5 surrogate (their stability regions are not identical), and (ii) the oscillatory eigenvalues of the system, where reliable simulation typically requires $\sim$20–30 steps per cycle. For our case, $f_d \approx 403$ Hz so $T = 1/f_d \approx 2.48$ ms; thus an accuracy-driven step is

$$h \lesssim \frac{T}{20\text{–}30} \;\approx\; 0.08\text{–}0.12 \text{ ms}.$$

By contrast, a theoretical $h_{\max}$ inferred from the Taylor-5 absolute-stability boundary is much larger (on the order of $\sim$0.5–1.0 ms in our plots), which explains the discrepancy with the empirically safe step used in simulation.

Question 7

Using a variable-time-step solver of your choice, establish a convergent solution that you can trust. Plot the solution i_L and v_C, and the time step h as functions of time. Your results should be accurate enough to be assumed as reference solution.

Discussion

After experimenting with all variable step solvers in MATLAB (ode45, ode23, ode113, ode15s, ode23s, ode23t, ode23tb), my findings are somewhat inconclusive. I personally liked stiff variations of ode23 (ode23s, ode23t, ode23tb). All subsequent comparison and discussion, including the reasoning behind my choice is in Q8.

Question 8

Using variable-step solvers, set $h_{max} = 1 \times 10^{-2}$ and both error tolerances $\varepsilon_{abs} = \varepsilon_{rel} = 10^{-3}$. Experiment with all the available variable-step solvers. In a table, print the maximum time-step size h_{max} chosen by each solver and the total number of steps taken. Comment on how these solutions compare with the reference solution found in part 7. Which solver in your opinion is more efficient and does the best job for this system? Compare the step-size and the overall number of steps observed in parts 6, 7, and this part.

Discussion

While nonstiff solvers (ode45, ode23) generally showed lesser error and had better timings, their transient response 'shape' was less consistent with a damped sinusoid as expected from the analytical solution. Stiff solvers (ode15s, ode23s) produced responses that visually aligned better with the expected damped sinusoidal shape, but they exhibited higher numerical error and longer computation times. Still, I personally found any stiff solver variation of ode23 (ode23s, ode23t, ode23tb) to be the best compromise between accuracy, speed, and expected response shape.

Upon closer inspection, the differences in solver performance can be attributed to their underlying numerical methods. Nonstiff solvers like ode45 and ode23 use explicit Runge-Kutta methods, which are efficient for problems with smooth solutions but may struggle with capturing the damping effects accurately in stiff systems. On the other hand, stiff solvers such as ode15s and ode23s employ implicit methods, which are better suited for handling rapid changes in dynamics but come at the cost of increased computational effort and potential numerical damping.

Among the stiff solvers, the variations of ode23 (ode23s, ode23t, ode23tb) stood out as a balanced choice. These solvers demonstrated a reasonable trade-off between computational efficiency and the ability to replicate the expected transient response shape. For instance, ode23s, which uses a modified Rosenbrock formula, was particularly effective in maintaining stability for larger time steps while preserving the qualitative features of the solution.

In summary, the choice of solver depends heavily on the specific requirements of the problem, such as the desired balance between accuracy, computational cost, and fidelity to the expected solution shape. All subsequent comparison and discussion is in Q8, where these findings are analyzed in greater detail.

Question 9

Question 10

Question 11

Question 12